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Summary

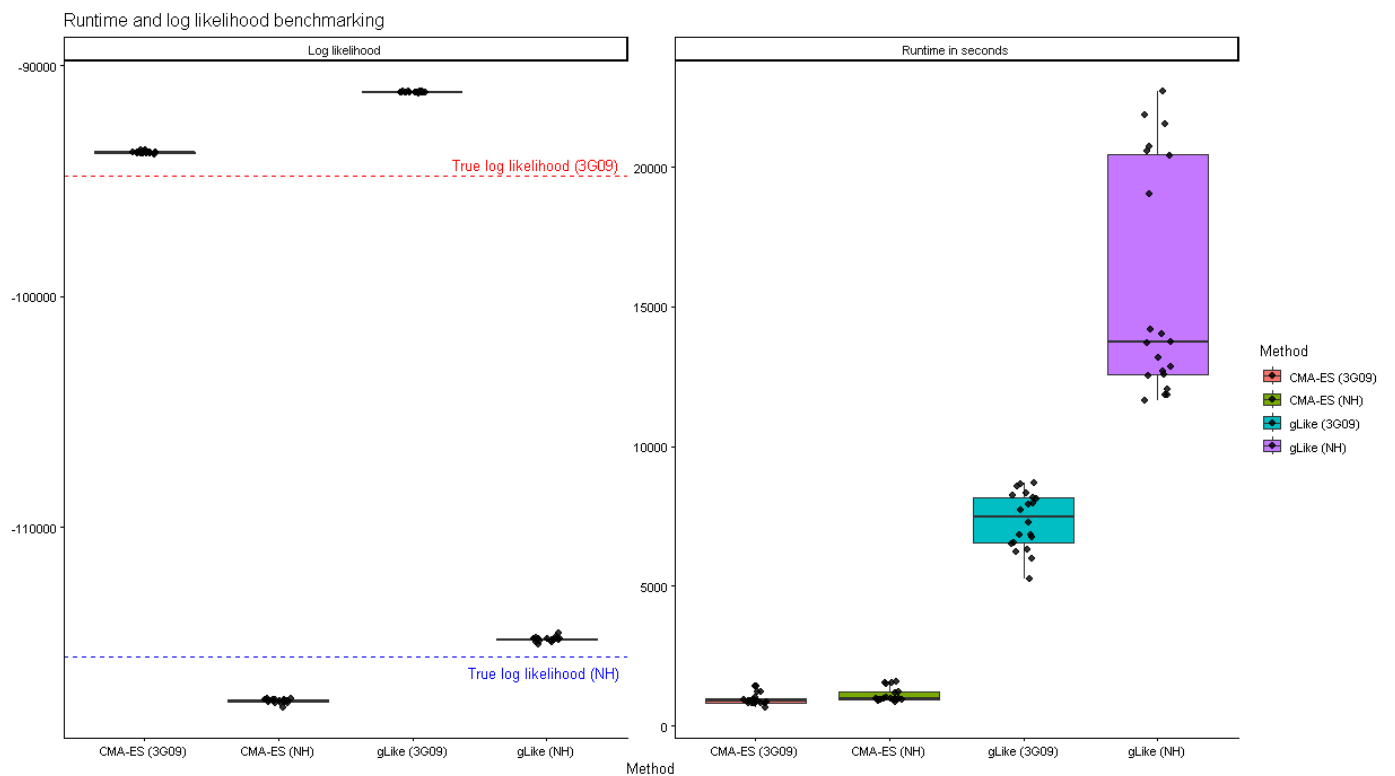
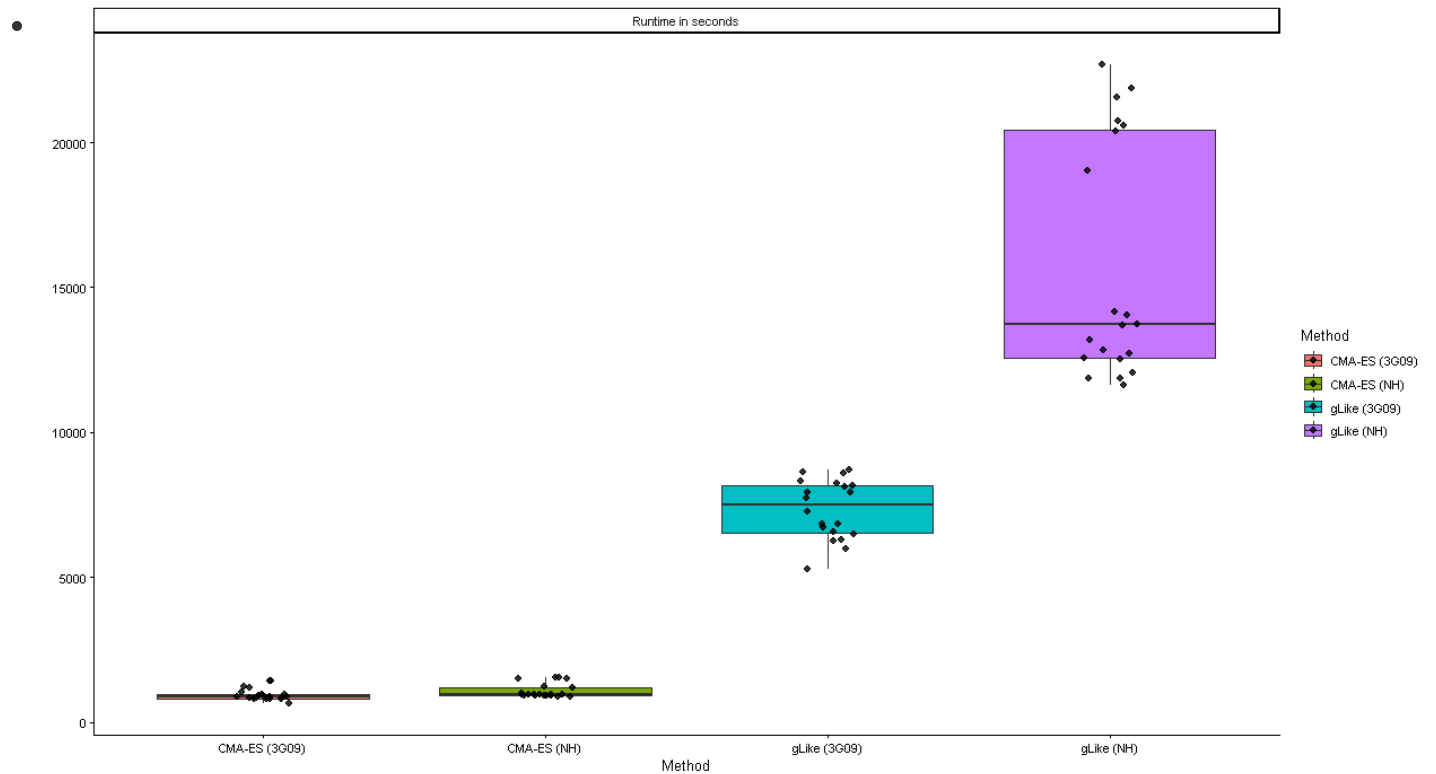
- gLike is a maximum likelihood framework for demographic inference using Ancestral Recombination Graphs
 - The currently implemented optimizer for gLike is fairly simple but straight forward. We iterate through each parameter, holding all others constant, and then step "up" or "down" in value towards a greater likelihood state. We repeat until we reach an optimum
 - Issues arise in terms of local vs. global optima due to the high dimensionality of the likelihood surface
 - Issues arise in terms of runtime, as we only change single parameters per iterative step
 - Additionally, we lack insight into correlative relationships between parameters
 - To address these issues I've implemented a different optimizer strategy: Covariance Matrix Adaptation Evolution Strategy (CMA-ES)
 - The basic premise is that we maintain a set of candidate solutions within some span or L2-norm vector distance of our current parameter state.
 - We sample the pool of candidates around us to iteratively compute a covariance matrix which gives insight into the local topology of the likelihood surface in a manner similar to computing a matrix of second derivatives (but not actually doing so).
 - We shift the current parameter state towards the direction indicated by the covariance matrices, and then shrink the span of the candidate pool
 - We keep repeating this process until we find ourself at an optimum
 - Hypothetically, due to the widespread sampling of candidates, we avoid issues arising from topological features (ridges / plateaus)
 - Hypothetically, due to the covariance matrices, we gain insight into correlative relationships between parameter pairs
 - Hypothetically, since we change multiple parameters at once per each iteration, we more effectively "spend" likelihood calculations

Quick note: (4A21)

- Runtime for gLike optimizer exceeded 12 hours and the job was autocanceled
- There is an implementation bug for this demographic model w.r.t CMA-ES (I need a more elegant solution)

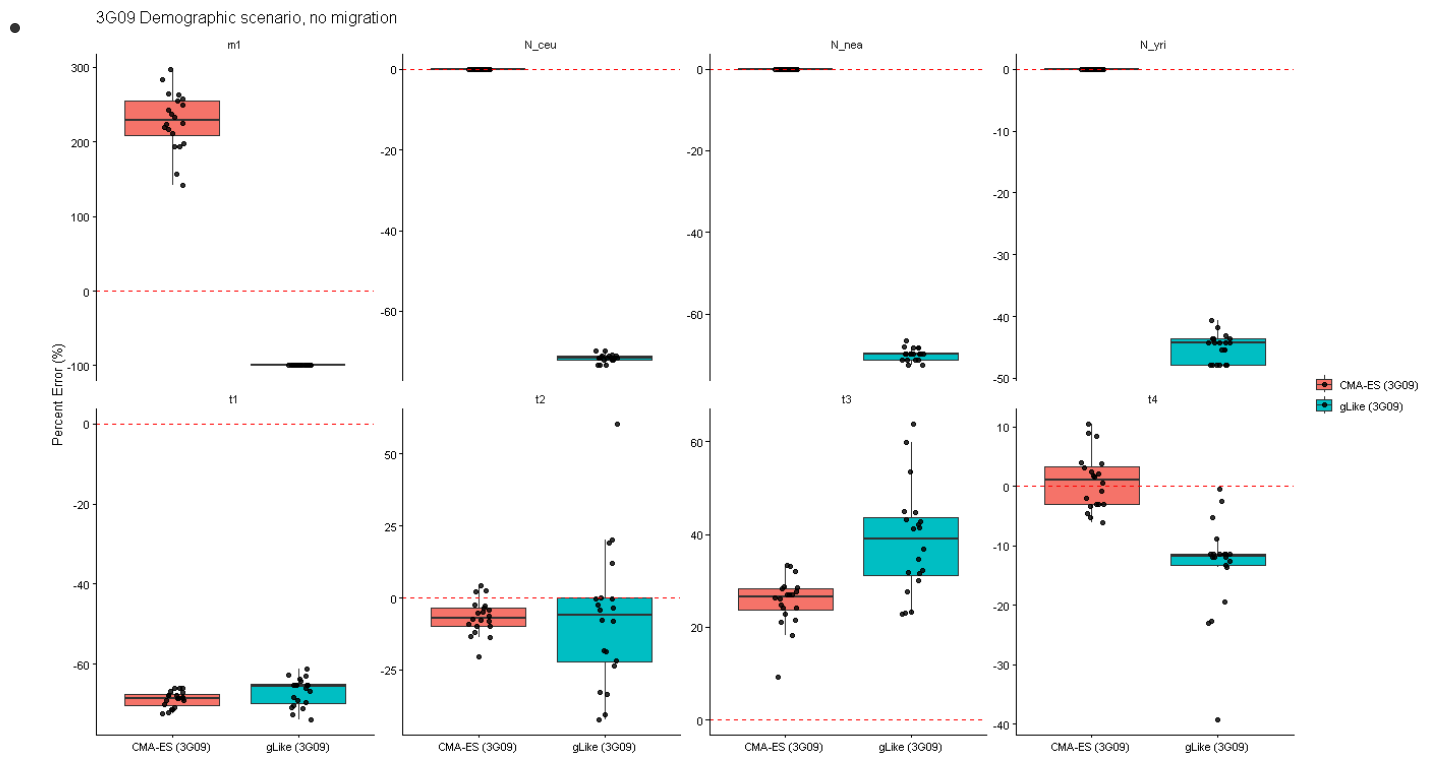
Benchmarking

Runtime and log likelihood

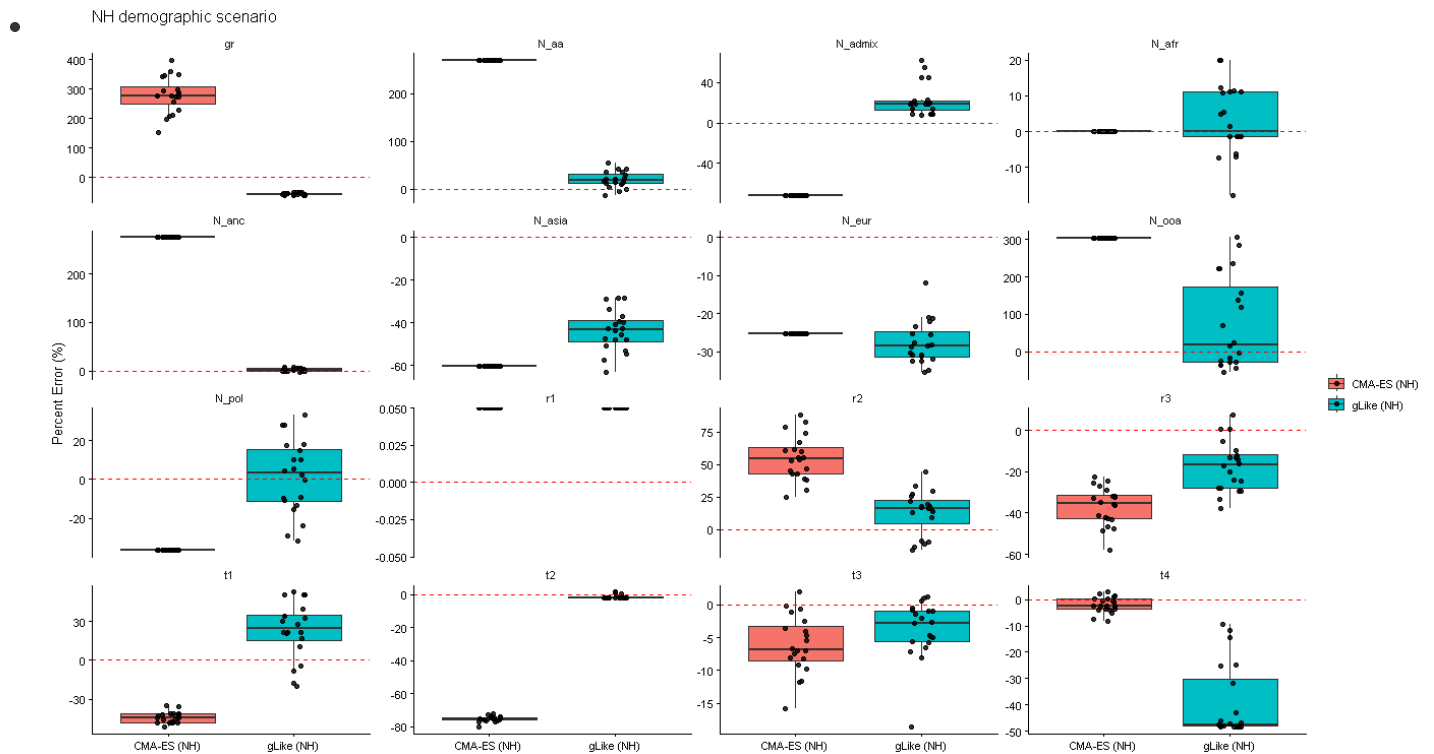


- Note that the log likelihood fit for the CMA-ES runs is **better** than the fit of the true underlying demographic parameters

Parameter fit percentage error (3G09)



Parameter fit percentage error (NH)



TODO

- Look into parameter scaling transform so that parameter magnitude doesn't interfere with

- Take three populations: 3G09, NH (or American Admixture), ancient European (see Caoqi's paper) [WIP]
- Diagnostic plots (talk to Ji Tang) which show parameter trajectory over time